GTRY164 Magnetic Keyboard Sensor and LED Driver

Chip Datasheet

V1.9

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1. General Description

GTRY164 is a low power chip for magnetic backlit keyboard application, which integrates a Hall sensor for magnetic switch and a RGBW LED Driver into one chip. The Hall sensor senses the distance of the magnet in the switch, and digitizes it into a digital value by the internal magnetic processing unit, and the digital sample value can be transmitted to the MCU through the 3-wire serial port.

With the 6rd generation Hall detecting technology, GTRY164 can realize up to 0.08 Gauss resolution and up to 32KHz sampling rate for full size keyboard. “With G.TECH's enhanced algorithm and appropriate switch, GTRY164 can detect 0.01mm relative resolution of the magnet displacement." GTRY164 achieves a sampling rate of 104-key keyboard less than 0.032ms. It uses constant current driver to drive RGBW LED, and this RGBW four-channel constant current value can be adjusted separately. Combined with the 13bit PWM, a very smooth and fine- tuned true color can be achieved.

Multiple 2-in-1 chips can be connected in series by daisy chain, and each chip parameter on the daisy chain can be separately configured through the 3-wire serial port, such as RGBW current and PWM parameters.

2. Features

- **Communication**
  - 3-wire serial port
  - Support daisy chain serial control
  - The magnetic induced voltage is quantized into a digital sample values
  - Support 32KHz reporting rate
  - Support both polling and interrupt-driven mode
- **PWM**
  - Supports 4 channel 13-bit constant current open-drain PWM output
  - Supports brightness adjustment with a 15 bit PWM cycle
  - Individual 17 coarse DC current steps,they are 0.6 / 0.9 / 1.2 / 1.8 / 2.4 / 2.7 / 3.0 / 3.6 / 4.2 4.5 / 4.8 / 5.4 / 6.0 / 6.3 / 7.2 / 8.1 / 9.0mA
  - 256 fine DC current adjustment for Individual Coarse current
  - Integrated high-efficiency LED drive technology in low duty cycle mode
- **Magnetic axis adaptability**
  - Up to 0.08 Gauss resolution
  - Input signal range supports±(20~300)mT configurable
- **Power supply**
  - Power supply voltage range of VDD：3.1~5.5V
  - Support sleep mode : Sleep power 1uA
  - Support SPS[smart power-saving] mode, which can reduce power consumption by 60%
  - Interrupt mode can reduce power consumption caused by communication by 90%.
- **ESD performance and temperature**
  - 2KV HBM
  - -10°C～+70°C
- **Package**
  - DFN2x2-8
- **Application**
  - USB RGB Magnetic Keyboard
  - 2.4GHz or BLE Magnetic Keyboard

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3. Functional Block Diagram

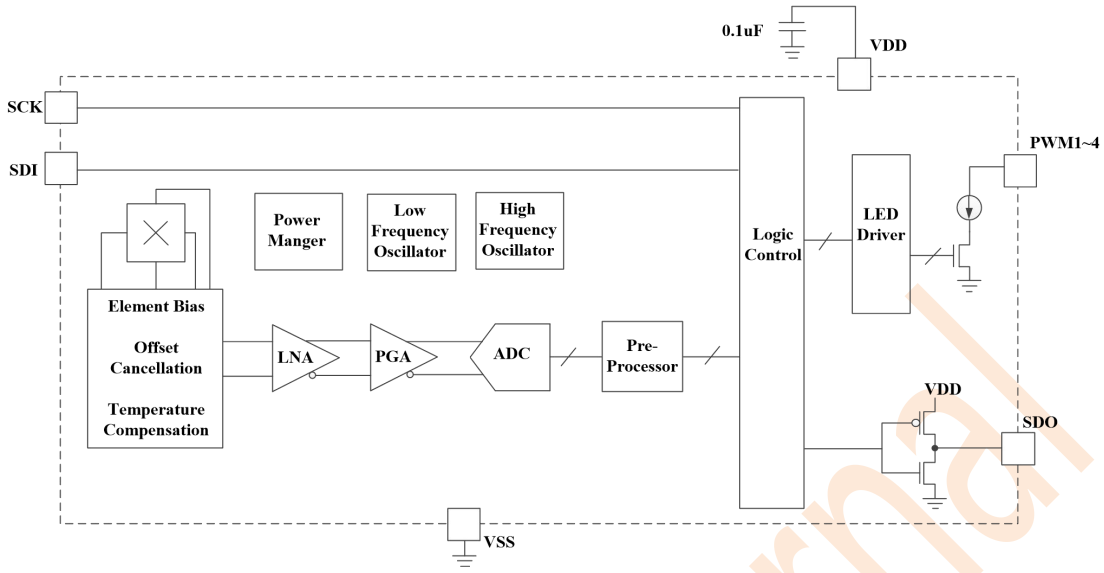


Fig 1. Functional Block

4. Pin Assignment

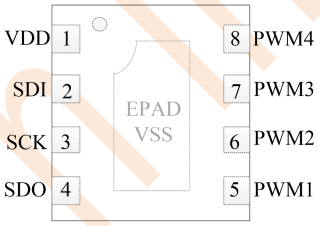


Fig 2. Top View

Table1 Pin description

Num	Name	I/O	Description	Remark
1	VDD	—	Supply voltage	
2	SDI	I	Serial data input	
3	SCK	I	Clock input	
4	SDO	O	Serial data output	
5	PWM1	O	PWM Constant current open drain output for LED4	
6	PWM2	O	PWM Constant current open drain output for LED3	
7	PWM3	O	PWM Constant current open drain output for LED2	
8	PWM4	O	PWM Constant current open drain output for LED1	
9	VSS	—	Ground EPAD	

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5. Electrical Characteristics

5.1 Absolute Maximum Ratings (at TA=25°C)

The working conditions of the chip exceed the following range, which will affect the reliability of the device.

Characteristics	Symbol	Rating	Unit
VDD Pin Voltage	V_{DD}	-0.3~6.0	V
Input voltage	V_{in}	-0.3~6.0	V
Output voltage	V_{out}	-0.3~6.0	V
Storage Temperature Range	T_{ST}	-40 ~ 85	℃
Operating Junction Temperature	T_J	115	℃
Operating Temperature Range	T_{OP}	-10 ~70	℃
Electrostatic Discharge Voltage	HBM	2.0	KV
	CDM	500	V
Steady State Life Test@85℃	-	3000	hrs
Welding Temperature	-	260 (<10s)	℃

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5.2 DC Electrical Characteristics

Characteristics	Symbol	Min	Typ	Max	Unit
Supply Voltage of VDD	V_{DD}	3.1	5.0	5.5	V
Supply Current @ Active	I_{Active}		460	530	uA
Supply Current @ Standby	$I_{Standby}$		20	30	uA
Supply Current @ Sleep	I_{Sleep}		1	2	uA
PWM Constant Current	I_{PWM}	0.01		9.0	mA
Current sink headroom voltage of LED Driver *[m=1.0 ,K=255,The current reaches 90%]*	V_{HR1}		0.50	0.60	V
Current sink headroom voltage of LED Driver *[m=1.0 ,K=255,The current reaches 97%]*	V_{HR2}		0.70	0.80	V
Current sink headroom voltage of LED Driver *[m=1.5 ,K=255,The current reaches 90%]*	V_{HR3}		0.70	0.80	V
Current sink headroom voltage of LED Driver *[m=1.5 ,K=255,The current reaches 97%]*	V_{HR4}		0.90	1.00	V
Input SMIT Hysteresis Voltage	V_{IHS}	0.50			V
Low level input voltage	V_{IL}			$0.2*V_{DD}$	V
High level input voltage	V_{IH}	$0.8*V_{DD}$			V
Low level output voltage	V_{OL}			$0.2*V_{DD}$	V
High level output voltage	V_{OH}	$0.8*V_{DD}$			V
SCK/SDI Input Capacitor	Cin			5	pF
Quiescent Output Error @LNA Gain=16	QOE16	-500	0	+500	LSB
Quiescent Output Error @LNA Gain=24	QOE24	-700	0	+700	LSB
Quiescent Output Error @LNA Gain=48	QOE48	-850	0	+850	LSB
Quiescent Output Error @LNA Gain=96	QOE96	-1000	0	+1000	LSB
Input magnetic field range	I_m	-300		300	mT

Note :

1) VDD = 3.3V, TA = +25°C, unless otherwise noted;
2) When I_{Active}, Hall sensor & ADC enable, Hall scan rate is 8KHz ;
3) when $I_{Standby}$, Hall sensor & ADC disable, OSC & LDO enable; For applications that only drive LEDs without considering button detection.
4) when I_{Sleep}, Hall sensor、ADC、OSC、LDO disable;
5) I_{PWM} is LED constant current and it can be adjusted within the range of 0.01~9.0mA .

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5.3 SPI Electrical Characteristics

IC Support 4-wire slave SPI，Supports full duplex data transmission with a data bit width of 8 bits。

Table2. SPI Interface Description

Num	SPISignal name	IC Signal name	IC Pin direction	Description
1	SPI_SS	-	-	Default enable, no chip selection signal
2	SPI_CLK	SCK	INPUT	SPI Clock
3	SPI_MOSI	SDI	INPUT	SP MOSI
4	SPI_MISO	SDO	OUTPUT	SPI MISO

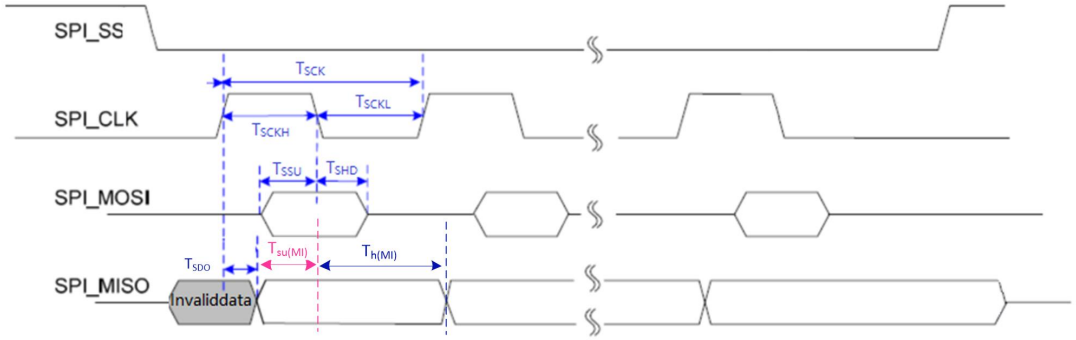


Fig 3. SPI Timing Diagram

Table3. SPI timing parameters

Symbol	Min	Max	Unit	Description
T_{SCKH}	20		ns	SPI_CLK High Level time
T_{SCKL}	20		ns	SPI_CLK Low Level time
T_{SCK}	40	1000	ns	SPI_CLK frequency range is 1.0~25MHz
T_{SSU}	10		ns	SPI_MOSI Setup time
T_{SHD}	10		ns	SPI_MOSI Hold time
T_{SDO}		18	ns	The time required for SDO to output valid data
$T_{su(MI)}$	10		ns	The setup time required to return data on the host side
$T_{h(MI)}$	4		ns	The hold on time required to return data on the host side

Note:

1、Usually, the SPI signal provided by the MCU can meet the timing requirements. However, due to the long data transmission link, the relationship between the data SDO returned and the clock needs to be confirmed, as shown in the red label in the figure above. SPI_MISO needs to meet the $T_{su\ (MI)}$ timing requirements, otherwise it will cause SPI communication errors. Please refer to the selected MCU chip for the specific value of $T_{su\ (MI)}$.

2、The detection point of $T_{su\ (MI)}$ oscilloscope testing should be as close as possible to the MCU controller pins.

4、The maximum value of T_{SDO} , requiring the SDO load capacitance to be less than 15pF.

6. Typical Application mode

Connectivity	Mode	Sensor Scan Rate	Latency (us)	Hall-6 Power(uA)	Remark
Wired	Active 2	32KHz	32	1616	Continuous conversion, Support interrupt mode
Wired	Active 3	16KHz	64	874	
Wired/wireless	Active 4	8KHz	125	460	
Wired/wireless	Active 5	4KHz	250	220	
Wired/wireless	Active 6	1KHz	1000	69	
Wired/wireless	Active 7	133Hz	7500	26	
Wired/wireless	Idle mode1	4Hz	250,000	20	
Wired/wireless	Idle mode2	4Hz	250,000	1.2	MCU actively sleep and wake up H6
Wired/wireless	Sleep mode	-	-	1	

Note :

1) The data in the table does not include LED power consumption;
2) High speed mode can enable SPS , further reducing power consumption;

7. Communication Protocol

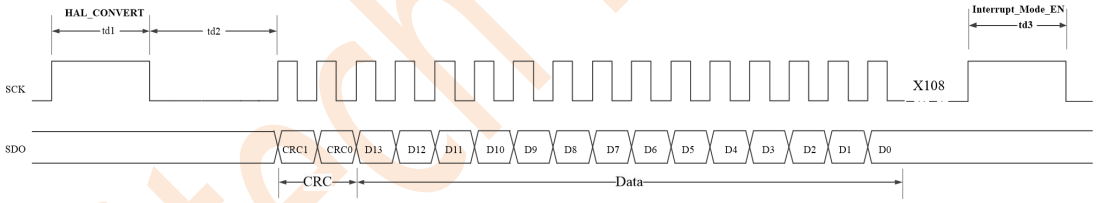
This GTRY164 uses only one SCK line to implement the communication protocol, and recognizes the instruction by calculating the high-level time of SCK and performs the corresponding action. Since SCK is also a register shift clock, in order to distinguish the instruction protocol from the normal read/write clock, the frequency of SCK as a clock must be higher than 1MHz. The read and write data is compatible with the SPI data format and supports only rising edge output data and falling edge sample data configuration options.

The data written or read is MSB first and LSB last.

7.1 Instruction Table

Num	Instruction Name	Description	Remark
1	HAL_CONVERT	Read buttons	
2	Interrupt_Mode_EN	Interrupt mode enable	
3	W_PWM_REG	Write PWM data	
4	SLEEP/WAKEUP	Sleep/Wake-up instruction	
5	WR_REG	End read and write register instruction	
6	FREQ_Calibrate	Oscillator frequency calibrate	
7	Soft_RST	Soft rest	

7.2 Instruction for HAL_CONVERT and Interrupt_Mode_EN



Hall conversion is 16bit data back mode. the conversion is shown in Figure 4:

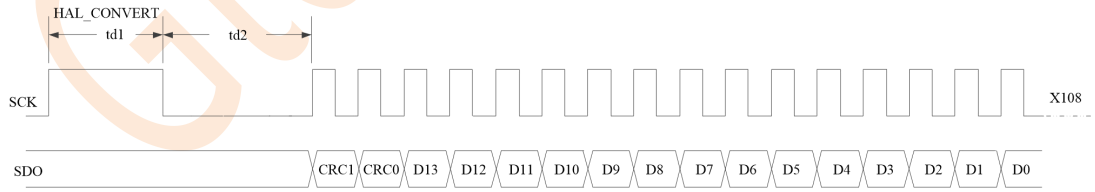


Fig 4. 16bit conversion mode

The returned data is fixed 16bit,the upper two bits CRC1/CRC0 are parity bits, D13~ D0 are data bits. The relationship between the parity bits CRC1/CRC0 and the data bits is as follows, where "^" stands for "exclusive OR":

CRC0=D0^D2^D4^D6^D8^D10^D12

CRC1=D1^D3^D5^D7^D9^D11^D13

There are two ways to enable interrupt mode:

One is to send the instruction td3 that enables interrupt mode; The second is to set the SCK idle timeout enable interrupt mode, as detailed in register 0x1B.

The instruction td3 is not necessary, nor does it have to be sent after reading key data. It can be sent at any time

[Note]: When the return data is 0xC000 or 0x3FFF, it indicates that the chip status is abnormal or communication is interfered , and resulting in unreliable communication data,the chip needs to be reset !

7.3 Instruction for write PWM

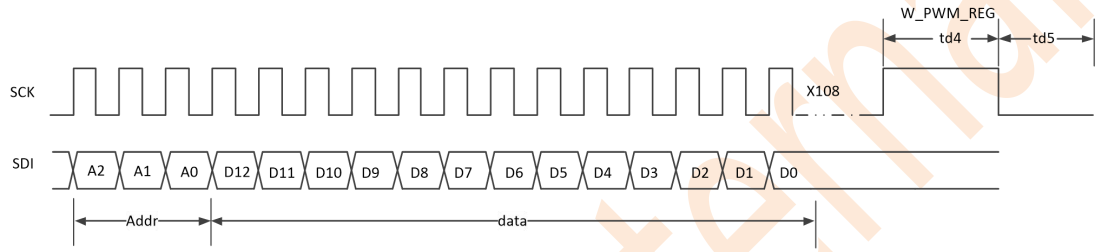


Fig 5. write PWM Instruction

The PWM register is completed by a write operation instruction. The instruction is as shown in the figure above. First, move the 3bit address and then 13bit data into the shift chain, and then send the PWM register instruction td3, wait for td5 to complete the write operation to the PWM register. The meaning of 16bit data is as follows:

Bit[15:13]: A2/A1/A0

① 3`b100：Write [D12:D0] 13bit PWM Cycle for all LED
② 3`b011：Write [D7:D0] PWM4 Duty for LED4 , [D8:D12] must be equal to zero
③ 3`b010：Write [D7:D0] PWM3 Duty for LED3 , [D8:D12] must be equal to zero
④ 3`b001：Write [D7:D0] PWM2 Duty for LED2 , [D8:D12] must be equal to zero
⑤ 3`b000：Write [D7:D0] PWM1 Duty for LED1 , [D8:D12] must be equal to zero

7.4 Instruction for WR_REG

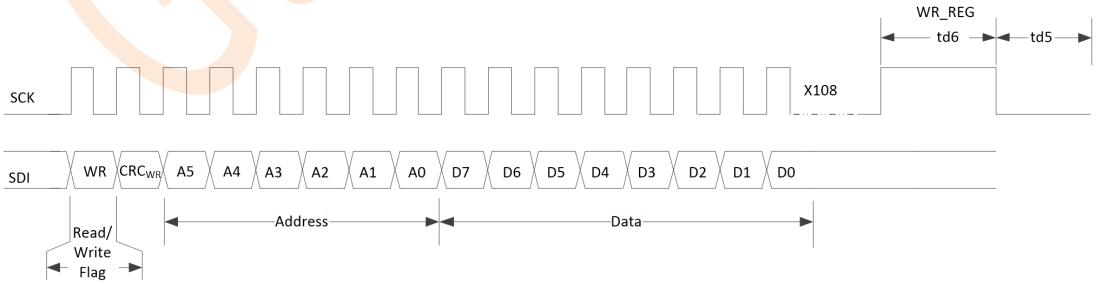


Fig 6. Write Register Instruction

The operation of writing the register is as shown in Fig 6. The write operation of the register requires one instructions to be executed. First, the 1-bit read/write flag bit "1" and 1-bit CRC_{WR},and the 6-bit address are shifted into the shift chain through the SPI-like interface timing.

8-bit data, and then send and write end command to complete the write register operation.

$$CRC_{WR}=A5\text{^}A4\text{^}A3\text{^}A2\text{^}A1\text{^}A0\text{^}D7\text{^}D6\text{^}D5\text{^}D4\text{^}D3\text{^}D2\text{^}D1\text{^}D0$$

Note: if the value of CRC_{WR} is incorrect,the instruction is invalid.

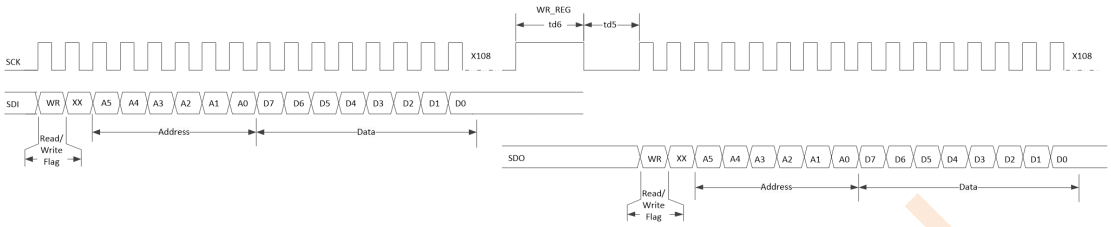


Fig 7. Read Register Instruction

The operation of reading the register is as shown in Fig7. The completion of a read register operation also requires one instructions to complete, first write the 1-bit read/write flag bit “0”, 7-bit address, and 8-bit data through the SPI-like interface timing，and then the 8bit data can be any value. then send the end read and write register instructions, and finally read the data back from SDO through the SPI-like interface timing, read back the data is 16bit, of which the lower 8bit is valid data.

7.5 Instruction for SLEEP/WAKEUP



Fig 8. Sleep Instruction

The sleep instruction can put the chip into a low-power sleep state. The instruction is as shown in Fig8. After the SLEEP instruction is sent, the chip goes to sleep. During the sleep, SCK needs to remain at a low level. When SCK is pulled high and td12 is maintained, the chip will be waked up, and it can work after td13.

7.6 Instruction for FREQ_Calibrate

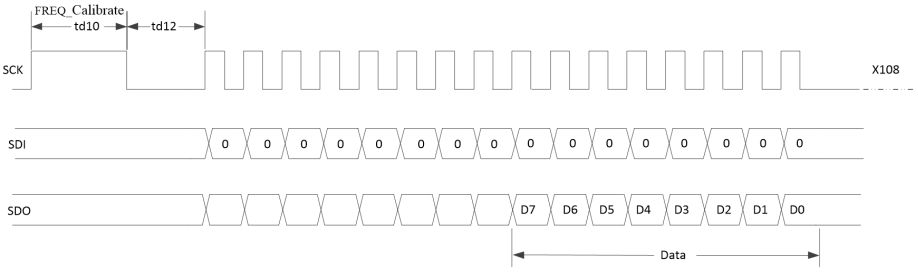


Fig 9. FREQ_Calibrate Instruction

This instruction is used to Calibration the internal oscillator frequency of the Hall chip. The instruction is as shown in Fig9. After sending the instruction, waiting for the fixed delay td10, the frequency value can be read back through the SPI-like interface timing, and the read back data is 16bit, where the lower 8 bits are valid data。

7.7 Instruction for Soft_RST

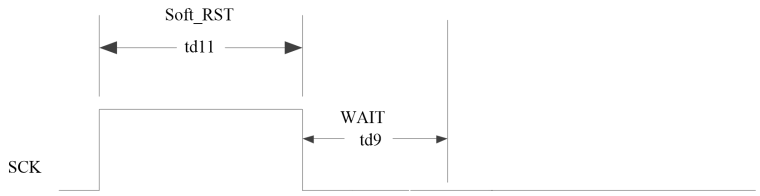


Fig 10. Soft_RST Instruction

This instruction is used to reset the Hall chip. The instruction is as shown in Fig10. After sending the instruction, waiting for the fixed delay td9. After the chip resets, it must be reinitialized to function properly.

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7.8 Timing table of instruction

Num	Symbol	Timing (Unit : us)	Remark
1	td1	5 ± 5%	Hall Convert
2	td2	Continuous mode : > 4 Single mode: > 24	Hall Convert Wait time
3	td3	12 ± 5%	Interrupt_Mode_EN
4	td4	18 ± 5%	Write PWM
5	td5	> 3	WR_REG & Write PWM Wait time
6	td6	39 ± 5%	WR_REG
7	td7	28 ± 5%	SLEEP
8	td8	2 ± 5%	WAKEUP
9	td9	> 50	WAKEUP & Soft reset Wait time
10	td10	128 ± 5%	FREQ_Calibrate
11	td11	>1280	Soft reset
12	td12	> 50	FREQ_Calibrate Wait time

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8. Register Description

The all register address is 6bit,bit width is 8bit:

Name	Addr	WR	Description	Default Value
LED_Current_1	0x5	WR	LED1 DC Current configure value k1 $I_{LED1} = I_{max}*m*k1/255$	0x0
LED_Current_2	0x6	WR	LED2 DC Current configure value k2 $I_{LED2} = I_{max}*m*k2/255$	0x0
LED_Current_3	0x7	WR	LED3 DC Current configure value k3 $I_{LED3} = I_{max}*m*k3/255$	0x0
LED_Current_4	0x8	WR	LED4 DC Current configure value k4 $I_{LED4} = I_{max}*m*k4/255$	0x0
LED_IMAX	0x09	RW	[7:6] : Reserve [5]: LNA Gain Control, See Reg 0xB for details [4]: LED_Multiplier : m 1: m=1.5 0: m=1.0 [3:0]: LED Imax configure ---------- I_{max} 4bit configure details as follows : 0x0: 0.0mA [Default] 0x1: 0.6mA 0x2: 1.2mA 0x3/4: 1.8mA 0x5/8: 2.4mA 0x6/9: 3.0mA 0x7/A: 3.6mA 0xB/C: 4.2mA 0xD: 4.8mA 0xE: 5.4mA 0xF: 6.0mA	0x20
Gain_Config0	0xB	WR	[7:4]: LNA Gain Configure 0xA & Reg 0x9 Bit5=1 : 96x gain 0x9 & Reg 0x9 Bit5=1 : 48x gain 0x8 & Reg 0x9 Bit5=1 : 24x gain [Typical] 0x8 & Reg 0x9 Bit5=0 : 16x gain other : Reserve [3:0]: PGA Gain Configure 0x9 : 24.0x gain 0x8 : 22.8x gain 0x7 : 21.6x gain	0x15

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Name	Addr	WR	Description	Default Value
			0x6 : 20.4x gain 0x5 : 19.2x gain [Typical] 0x4 : 18.0x gain 0x3 : 16.8x gain 0x2 : 15.6x gain 0x1 : 14.4x gain 0x0 : 13.2x gain other : Reserve	
Test_Reg0	0xC	WR	Write fix value 0x83	0xAA
Sample_Interval_L	0xF	WR	[7:0]: Low 8bit of the ‘Sample_Interval	0x20
Sample_Interval_H	0x10	WR	[7:4] : Write fix value 0x2 [3:0] : High 4bit of the ‘Sample_Interval’ Fine configure interval time between two convert, 28~16384us configurable with a step of 4us Note: (1)Time of Sample_Interval= Sample_Interval*4 us (2)Unable to configure 0x0~0x6	0x20
Sampling_Interval_SPS	0x11	WR	[7:0] : Sampling_Interval_SPS Interval time between two convert in SPS mode, 0~255ms configurable with a step of 1ms Note:Final Sampling interval as follow Sampling Interval when the key is pressed =Sample_Interval [us] Sampling Interval when the key is released =Sample_Interval_SPS+1 [ms]	0x0
Key_Released_Timeout	0x12	WR	[7:0]：Key_Released_Timeout In smart power-saving mode, when key released time exceeds ‘Key_Released_Timeout’, the chip enters power-saving mode, and the ‘Key_Released_Timeout’ can be configured from 1 to 256 seconds, with a step of 1 second	0x3C

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Name	Addr	WR	Description	Default Value
			Time of Key_Released_Timeout = Key_Released_Timeout+1 s	
SCK_Idle_Time	0x1B	WR	[7:5]：Write fix value 0 [4:0]：SCK_Idle_Time In interrupt mode, the SCK low-level duration detection value determines when an interrupt trigger mode is activated if SCK remains low for longer than this value. SCK_Idle_Time=(SCK_Idle_Time+3)/2 unit：us	0x7
Test_Reg1	0x26	WR	Write fix value 0xA	0xA
Test_Reg2	0x27	WR	Write fix value 0x5	0x5
Test_Reg3	0x28	WR	Write fix value 0x4	0x4
Gain_Config1	0x29	WR	Linkage with LNA gain，Need to be configure： 0x50, when LNA Gain is 16x 0x60, when LNA Gain is 24x 0xA0, when LNA Gain is 48x 0xF0, when LNA Gain is 96x	0x40
Gain_Config2	0x2A	WR	Linkage with LNA gain，Need to be configure： 0x20, when LNA Gain is 16x 0x28, when LNA Gain is 24x 0x40, when LNA Gain is 48x 0x60, when LNA Gain is 96x	0x32
Press_Key_Threshold_L	0x2B	WR	[7:0]: Low 8bit of the ‘Press_Key_Threshold’	0xDC
Press_Key_Threshold_H	0x2C	WR	[7]:Reserve [6]: Key_Type1 for SPS mode and Interrupt_Trigger mode 1: When the key is pressed, the ‘ADC value’ increases; ‘ADC value’ large than the ‘Press_Key_Threshold’, indicates the key is pressed 0: When the key is pressed, the ‘ADC value’ decreases,‘ADC value’ less than the ‘Press_Key_Threshold’ ,indicates the key is pressed [5:0]: High 6bit of the ‘Press_Key_Threshold’	0x1E
Baseline_Initial_	0x2D	WR	[7:0]: Low 8bit of the ‘Baseline_Initial_Value’	0x00

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Name	Addr	WR	Description	Default Value
Value_L			**Note:** This chip only supports top baseline operations. All subsequent references to "Baseline" refer exclusively to the top baseline.	
Baseline_Initial_Value_H	0x2E	WR	[7:6]: Reserve [5:0]: High 6bit of the 'Baseline_Initial_Value' Baseline_Initial_Value Writing method: Step 1: Write 0x2D Step 2: Write 0x2E 1) The default is 8192; 2) When BASL_INIEN = 1, the Baseline_Initial_Value will be automatically updated with real-time ADC values,until BASL_INIEN = 0.	0x20
Test_Reg4	0x2F	WR	Write fix value 0x14	0x14
Baseline_Positive_Deviation	0x30	WR	[7:0]: BASL_PDEL Baseline tracking effective value positive deviation configuration value, sampling value less than [BASL_INI+16*BASL_PDEL] participates in Baseline operation, and the default is 512;	0x20
Baseline_negative_deviation	0x31	WR	[7:0]: BASL_NDEL Baseline tracking effective value negative deviation configuration value, sampling value large than [BASL_INI-16*BASL_NDEL] participates in Baseline operation	0x20
Key_Type	0x32	WR	[7]: Input magnetic field polarity reversal @1 [6]: Key type2 1: When the key is pressed, the 'ADC value' increases.; 0: When the key is pressed, the 'ADC value' decreases. [5]: Write fix value 0 [4]: smart power-saving mode Enable @1 [3:0]: Write fix value 0x1	0x41

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Name	Addr	WR	Description	Default Value
Work_Mode	0x33	WR	[7]：Hall_Conversion_Mode 0：Single conversion 1：Continuous conversion Note: After switching conversion modes, the FW must wait 50 us before reading the Hall sensor conversion data. [6]：Write fix value 0. [5]：Interrupt_Mode_EN, Enable@1 [4]：Interrupt_Trigger_Method， 0：Command Trigger 1：SCK Low Level Idle Timeout Trigger [3]：First_Hall_Flag **In interrupt trigger mode**: the first Hall chip of each chain must be set 0 , and other chip must be set 1 **In other mode**: the all Hall chip of each chain must be set 0 ; [2]：BASL_EN, Baseline mode Enable@1 [1]：Write fix value 0. [0]：BASL_INIEN When BASL_INIEN = 1, the Baseline_Initial_Value will be automatically updated with real-time ADC values,until BASL_INIEN = 0	0x0
Test_Reg5	0x34	WR	Write fix value 0x3	0x03
SDO_Drive_Current	0x35	WR	[7:4]：Write fix value 0x2. [3]：Write fix value 0. [2:0]：SDO IO Driver current configure: 3'b000 : 2mA 3'b001 : 4mA 3'b011 : 8mA 3'b111 : 12mA	0x27
Baseline_Real-time_Value	0x3C	R	[13:0]：Baseline Real-time Value **Note:** When reading this register using the WR_REG command, the returned data is 16 bits, with the lower 14 bits being the valid data.	0x0

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9. Detailed Description

9.1 Initialization Process

After the IC is powered on, perform the following operations in sequence:

Step1: Correct the frequency to ensure that the chips on the daisy chain are kept in sync;

Step2: Configure the initialization parameters to ensure that the chip enters the normal working state.

The first step after power-on is to perform frequency correction to ensure that the chip on the link is synchronized. The calibration scheme is as shown in the figure below. If the calibration fails, it indicates that the chip is abnormal, and the return value can be used to determine the position of the abnormal chip in the daisy chain.

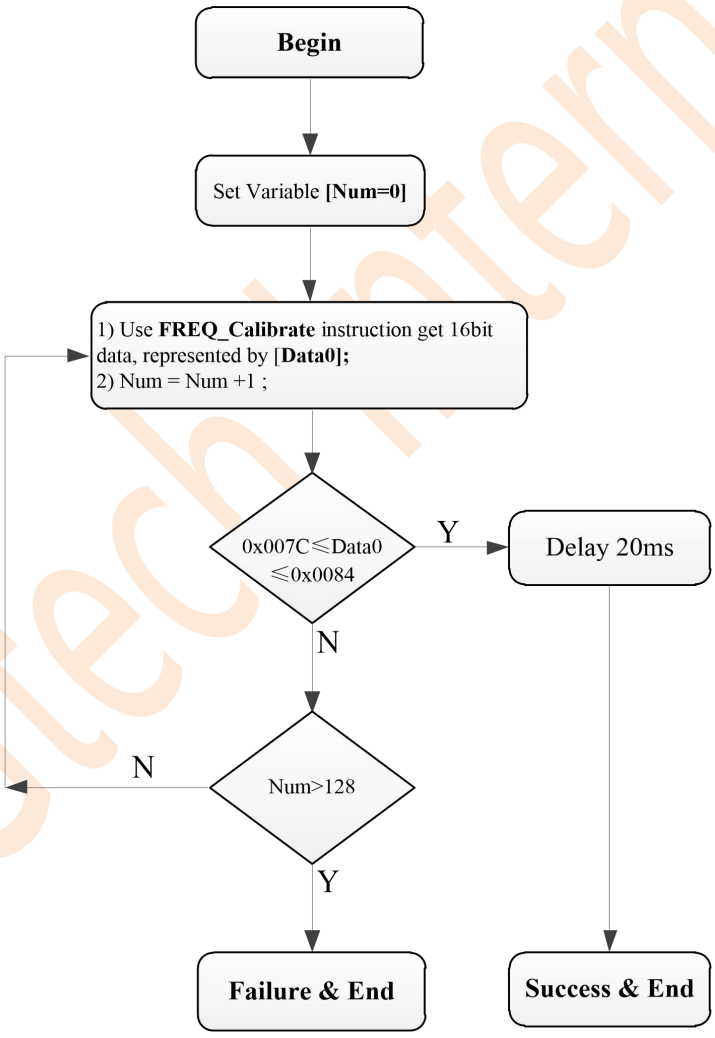


Fig 11. Frequency Calibration Procedure

Note: Each time a frequency instruction is issued, it is necessary to verify whether the return values of all keys satisfy the condition : 0x007C≤Data0≤0x0084.

9.2 Single Conversion Mode

Single conversion mode refers to the master MCU issuing a conversion command 'HAL_CONVERT', causing the chip to perform one ADC conversion, as shown in Fig 4. The MCU can initiate the conversion at any time, then wait for the duration of td2 before reading the button data.

9.3 Continuous Conversion Mode

In continuous conversion mode, the chip continuously cycles through conversions. The conversion rate can be set within the range of 60Hz to 32KHz, see register 0x10 for details.

The MCU can read data at any time without interrupting the Hall conversion process. The method for reading Hall data is identical to that in single conversion mode, as shown in Fig 4, but continuous conversion mode features a shorter td2 wait time.

9.4 Interrupt Mode

In interrupt mode, the chip continuously cycles through periodic conversions at a configurable rate between 60Hz and 32KHz, see register 0x10 for details. The MCU must define the key direction and the Hall change value Delta-A representing a key press，see register 0x12/0x2B/0x2C/0x32 for details. When the Hall sampling value change reaches Delta-A, a high-level interrupt signal is sent to the MCU via SDO until the Hall change value falls below Delta-A.

There are two ways to enter interrupt mode. The first is by issuing the Interrupt_Mode_EN command from the MCU, which can trigger the Hall chip into interrupt mode at any time. The second is through SCK idle timeout, where the Hall chip automatically enters interrupt mode when it detects the SCK signal remains low for a duration exceeding T0, which is configured with 1.5us~17.5us，see register 0x1B for details.

Hall data is read identically in interrupt mode as in continuous conversion mode, as shown in Fig 4. Interrupt mode significantly reduces EMI and dynamic power consumption caused by communication.

9.5 Sleep mode

Low power mode and wake-up need to follow the steps below:

1) Executed the SLEEP instruction to sleep;
2) Execute the WAKEUP instruction to wake up the chip;

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9.6 Smart power-saving mode [SPS]

In Smart Power-Saving Mode, the Hall chip configures different conversion rates by detecting whether the key is idle. Under normal conditions, the Hall chip operates at high speed. When it detects that the key has remained unchanged for longer than time T1, it reduces the Hall conversion rate, entering a low-speed conversion mode to lower power consumption.

The high-speed conversion rate (60Hz~32KHz), low-speed conversion rate (4Hz~1KHz), and idle time T1 (1~256s) can all be configured via the MCU, refer to registers 0x10/0x11/0x12 for details.

The parameter indicating key presence shares the same setting as the interrupt mode.

9.7 TOP Baseline

GTRY164 employs a specialized algorithm supporting TOP Baseline tracking to eliminate quantization value variations in the TOP position caused by factors such as key deformation, magnet aging, and temperature changes, thereby preventing false code transmission.

The TOP Baseline can be configured with an initial value and an effective change range, as detailed in registers 0x2D/0x2E/0x30/0x31. The MCU can read the real-time TOP Baseline value, as detailed in register 0x3C.

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9.8 LED Driver

The LED driver supports 4-channel constant-current operation, capable of driving RGBW LEDs with four distinct colors. The LED drive principle is illustrated below. I_{max} offers 10 selectable levels, with m configurable as either 1 or 1.5, and the parameters k1/k2/k3/k4 can be individually set within a range of 0 to 255. Combining these three parameters enables current selection from 0 to 9mA and white balance adjustment.

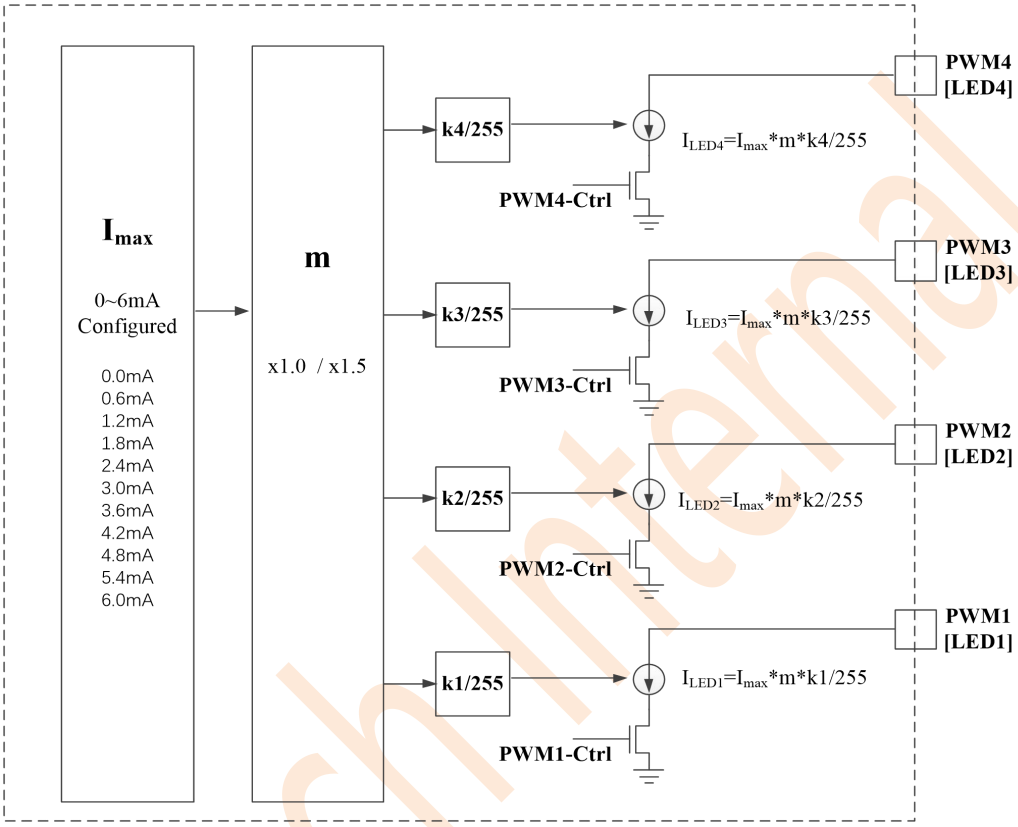


Fig 12. LED Driver Block

The 4-channel LED driver is fully equivalent, allowing any 3 channels to be selected for driving RGB LEDs. RGB color mixing is achieved via PWM, with the control method shown in Fig 5. For common color mixing, 8-bit PWM is used, mixing colors by adjusting the duty cycles of the 3 RGB PWM channels.

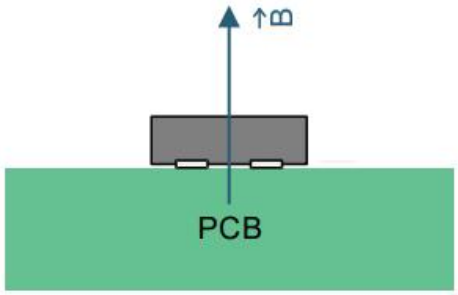
The GTRY164 also supports adjusting LED brightness by modifying the PWM cycle period. This period regulates all four PWM channels simultaneously, meaning all four LED brightness levels are adjusted concurrently. When using 8-bit PWM, the cycle period is 15 bits. PWM clock is 2MHz， the resulting LED current is calculated as follows:

1) I_{LED} = [I_{max} * m * k / 255] * [PWM_Duty / (PWM_Cycle - 2)]
2) PWM_Cycle value range: 257 to 8192 ;
3) When PWM_Duty=255 and PWM_Cycle=257, brightness is maximum, PWM frequency is 7.8KHz;
4) When PWM_Duty=255 and PWM_Cycle=8192, brightness is minimum, PWM frequency is 245Hz;

PWM control is implemented via dedicated instructions, detailed in Section 7.3.

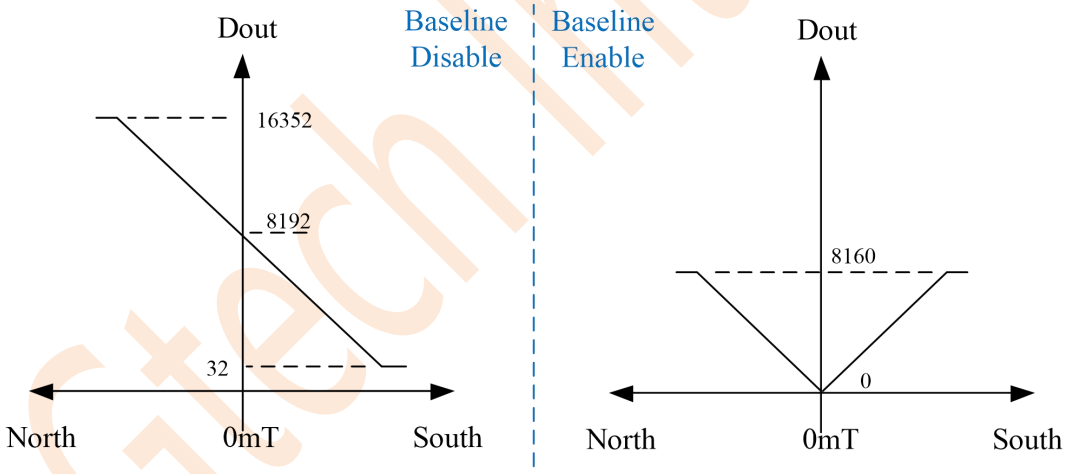
9.9 Magnetic Flux Direction

The device is sensitive to the magnetic field component that is perpendicular to the top of the package,direction of Sensitivity as shown in follow fig:



9.10 Correspondence between output and magnetic field

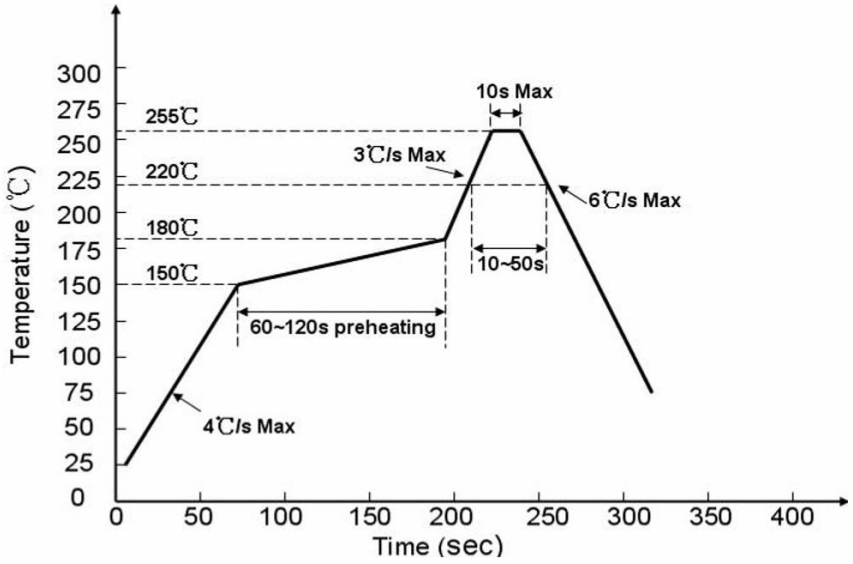
As shown in the figure below, the corresponding relationship between the magnetic field and the output, when the input magnetic field is 0mT (no magnetic field) and without considering the offset of hall chip, the output of the Hall sensor is 8192. When the south pole of the magnet is applied to the sensor, the output value of the sensor will decreases, and the decrease in the sensor output value changes linearly with the magnetic field strength until the sensor is saturated; correspondingly, when the north pole of the magnet is applied to the sensor, the sensor output value rises, and the increase in the sensor output value changes linearly with the S-type magnetic field strength until the sensor is saturated.



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10. Welding Requirements

10.1 SMT Reflow Soldering Instructions



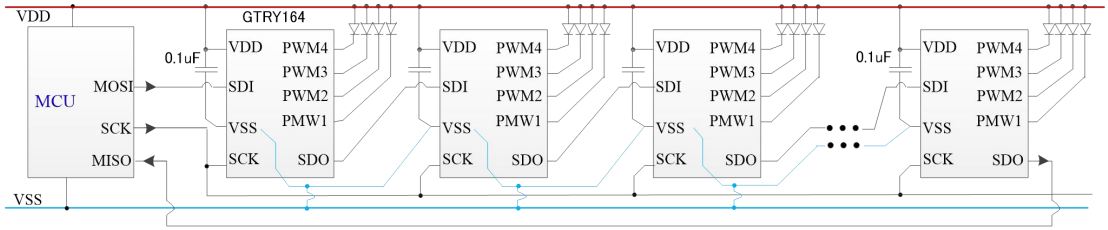
Note: It is recommended to use solder paste with high silver content and low particles, with an activity of 12 or more.

10.2 Hand Soldering Profile

The temperature of the iron should be lower than 350℃ and soldering within 3sec per solder-pad is to be observed.

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11. Reference Circuit



Note: Considering the reduction of EMI, it is strongly recommended to use 6~8 chains

12. Marking Information

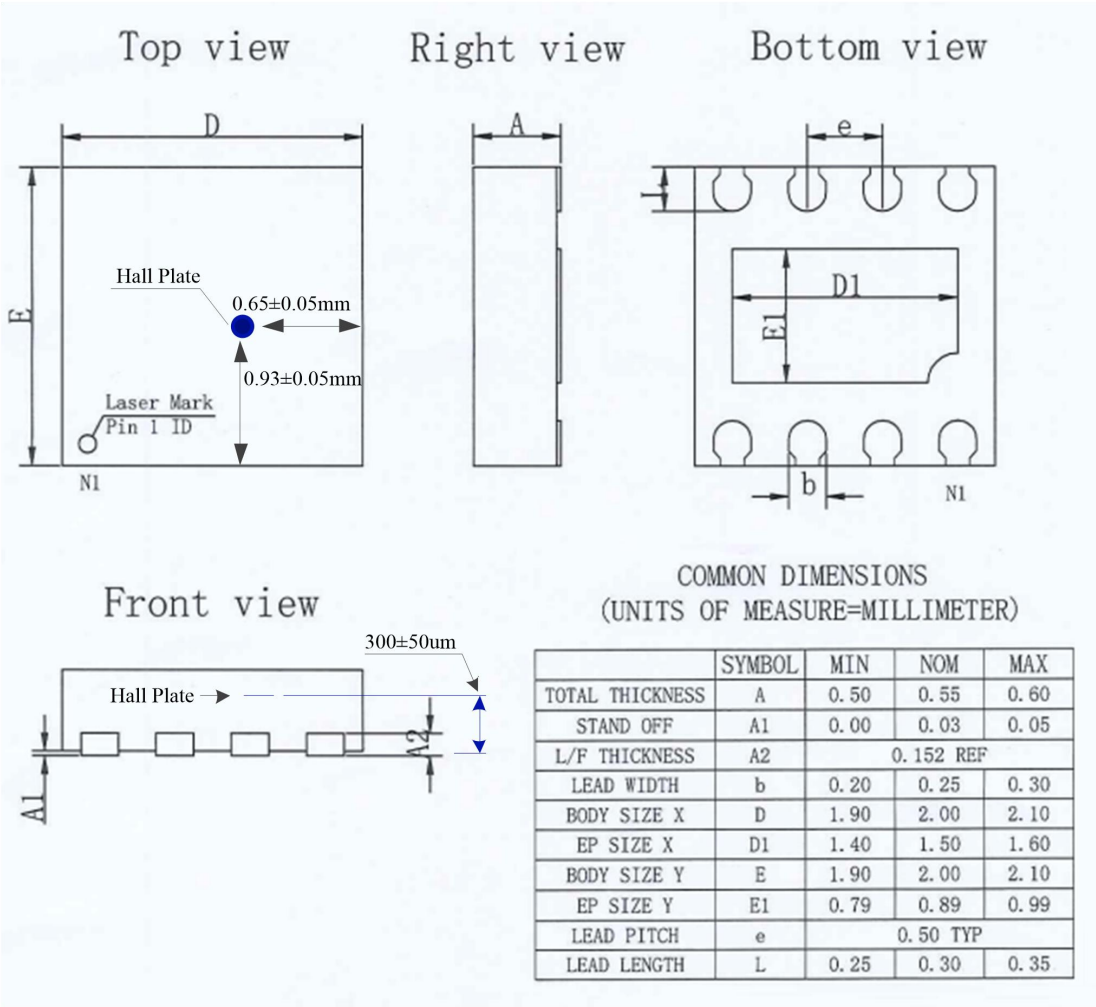
1) Marking:
① Line 1 is is product code : GTRY164
② Line 2 is the production batch number: YYWWXX. There may be hanges in this line, where YYWW is data code，YY is year，WW is week, and XX is the Serial Number

2) Examples of physical chips



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13. Package Outlines



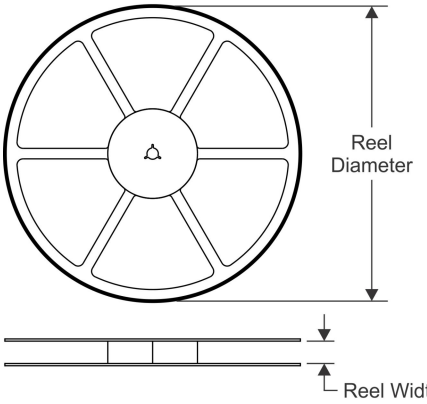
COMMON DIMENSIONS
(UNITS OF MEASURE=MILLIMETER)

	SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS	A	0.50	0.55	0.60
STAND OFF	A1	0.00	0.03	0.05
L/F THICKNESS	A2		0.152 REF	
LEAD WIDTH	b	0.20	0.25	0.30
BODY SIZE X	D	1.90	2.00	2.10
EP SIZE X	D1	1.40	1.50	1.60
BODY SIZE Y	E	1.90	2.00	2.10
EP SIZE Y	E1	0.79	0.89	0.99
LEAD PITCH	e		0.50 TYP	
LEAD LENGTH	L	0.25	0.30	0.35

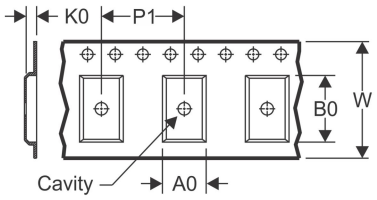
14. Package Materials Information

1. TAPE AND REEL INFORMATION

REEL DIMENSIONS

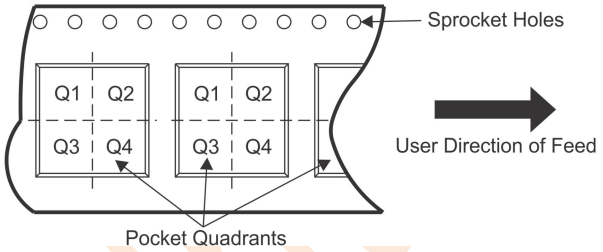


TAPE DIMENSIONS



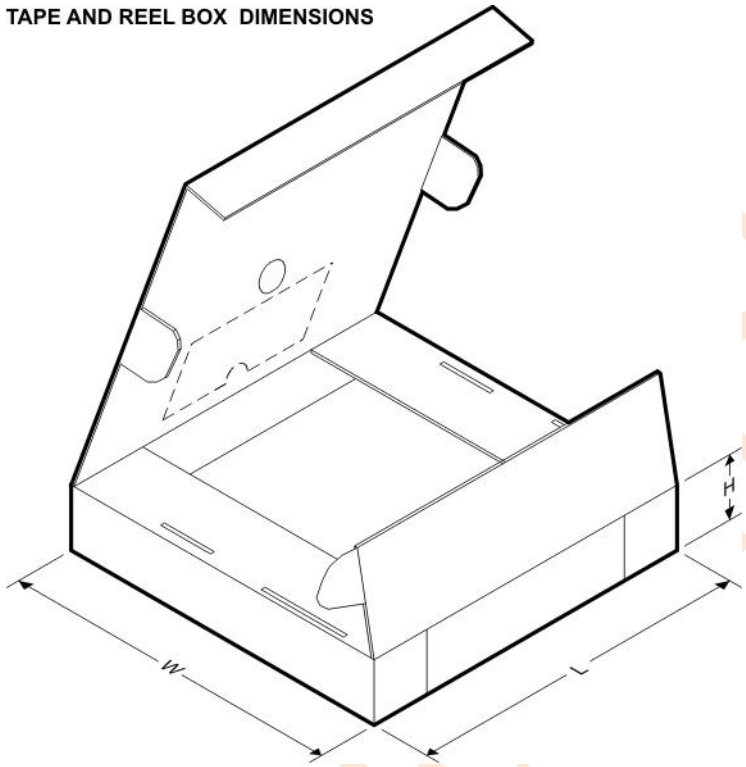
A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



Device Name	Package Type	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
GTRY164	DFN2x2-8	8	4000	330	12.5	6.40	5.3	2.01	8	12	Q1

2. TAPE AND REEL BOX DIMENSIONS



DeviceName	Package Type	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
GTRY164	DFN2x2-8	8	40000	357	340	50

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15. Summary of Changes

Rev	Paragraph(s) Changed	Description of Change(s)
V1.0		Initial version
V1.1	7.6 7.8 9	Modify the fig9, change td5 to td12; Add td12>50us , modify td11; 9.1 Modify Frequency Calibration Process
V1.2	9	Modify the description of the section
V1.3	5.2	Modify the 'Current sink headroom voltage of LED Driver'
V1.4	8	Modify Reg 0xC configure to 0x83
V1.5	5.3 8	Modify Table3 T_{SSU} & T_{SHD} Modify Reg 0x32 bit7
V1.6	5.3 7.8 8	Modify Table3 $T_{SU(MI)}$ Change td11 to 1280us Modify Reg 0x9/0xB for LNA Gain Control, Modify Reg 0x29/0x2A accordingly
V1.7	7.3 7.6 9.8	Revise the PWM control description Revise the fig.9 : SDI shall be 0 Follow 7.3, Revise the PWM control description
V1.8	5.2	Add 'Input SMIT Hysteresis Voltage'
V1.9	8	Modify Reg 0x33 bit[1]

Note: This document is an initial version, the specific parameter specifications are subject to the latest version of the manual,subject to change without notice

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